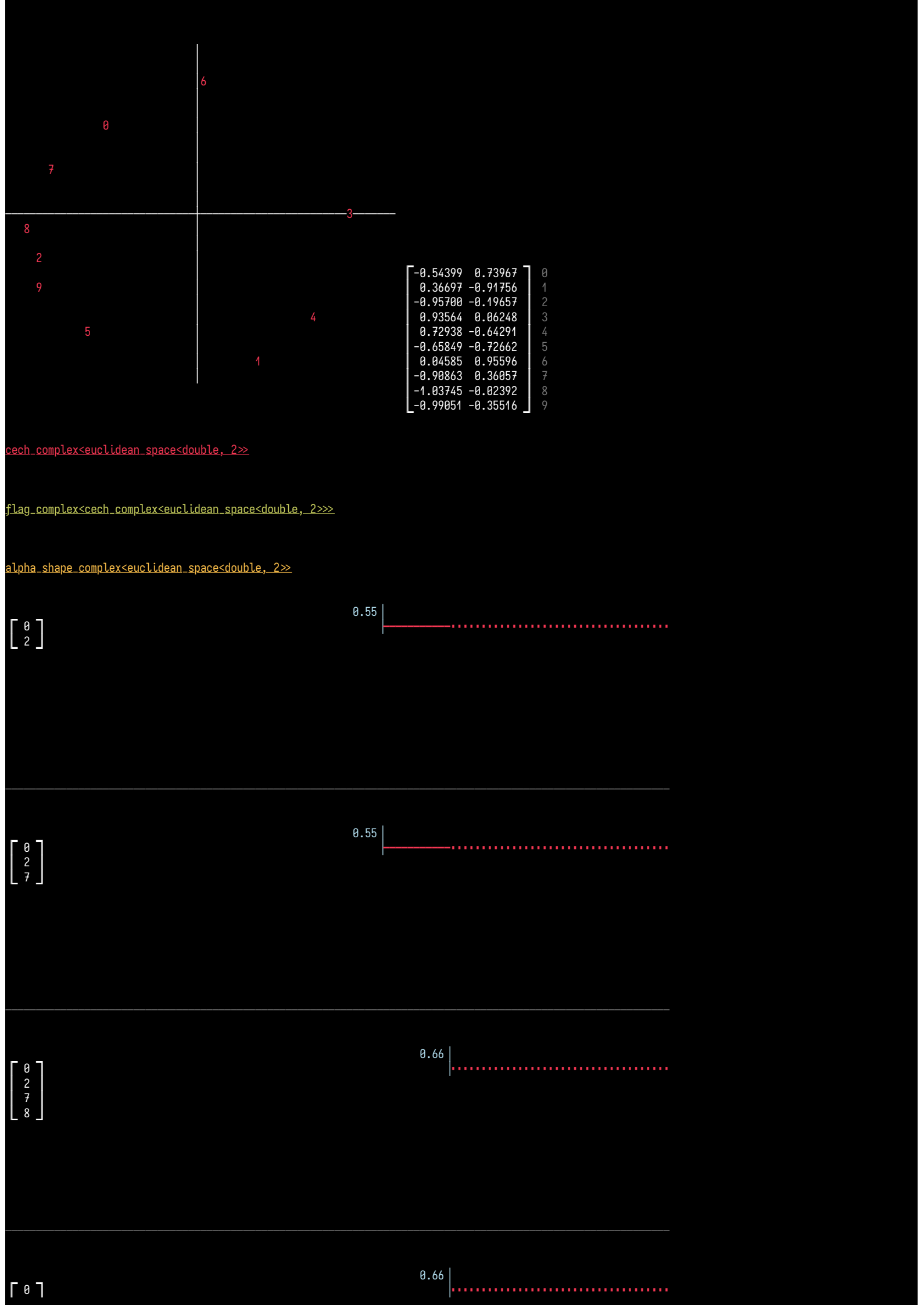




$$\begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

0.33

0.66

0.66

$$\begin{bmatrix} 0 \\ 6 \\ 7 \end{bmatrix}$$

0.61

$$\begin{bmatrix} 0 \\ 7 \end{bmatrix}$$

0.28

0.55

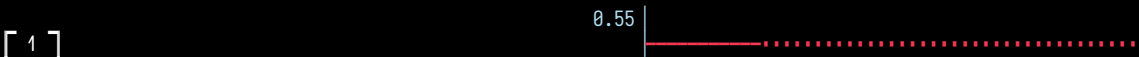
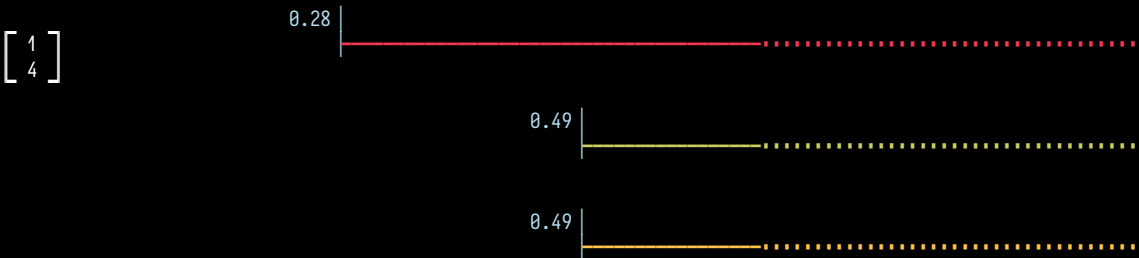
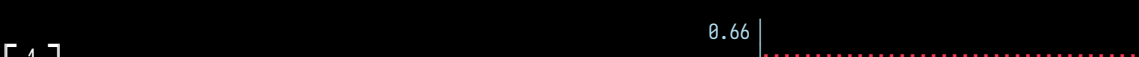
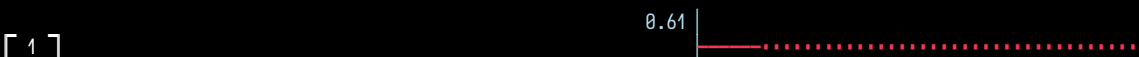
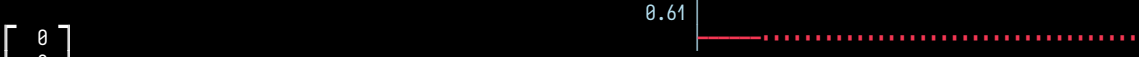
0.55

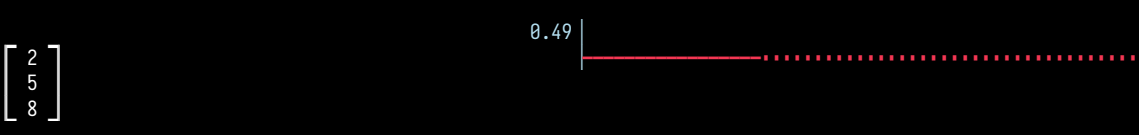
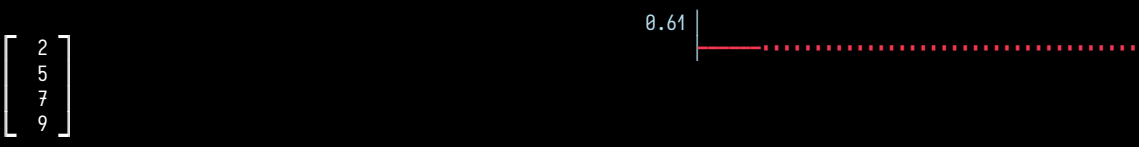
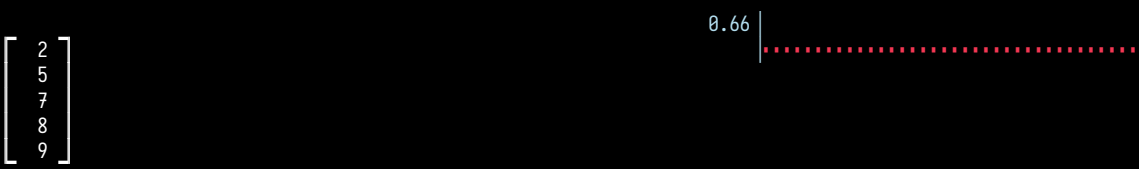
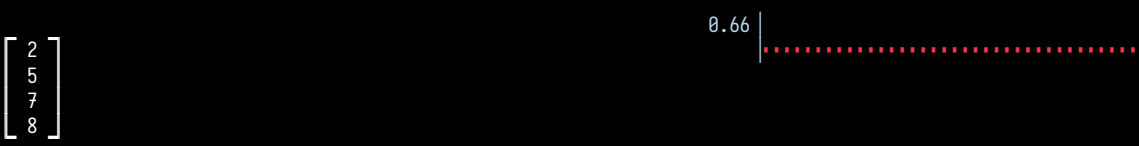
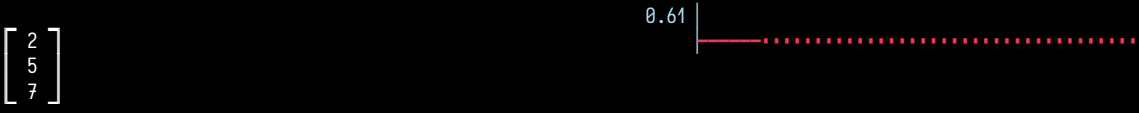
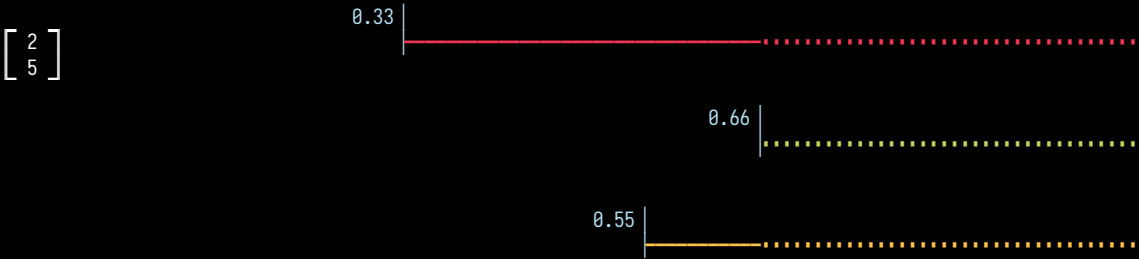
$$\begin{bmatrix} 0 \\ 7 \\ 8 \end{bmatrix}$$

0.49

$$\begin{bmatrix} 0 \\ 7 \\ 9 \end{bmatrix}$$

0.66





$\begin{bmatrix} 2 \\ 5 \\ 8 \\ 9 \end{bmatrix}$



$\begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}$



$\begin{bmatrix} 2 \\ 7 \end{bmatrix}$



$\begin{bmatrix} 2 \\ 7 \\ 8 \end{bmatrix}$

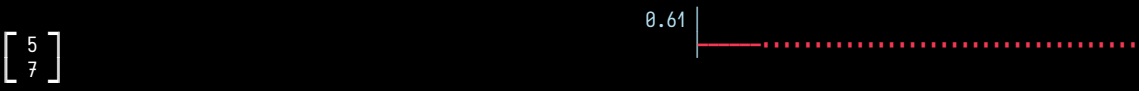
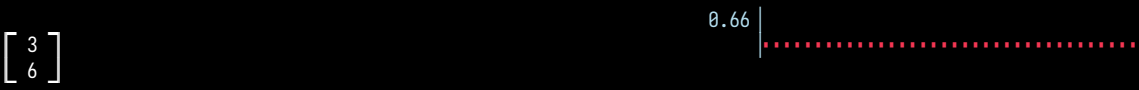
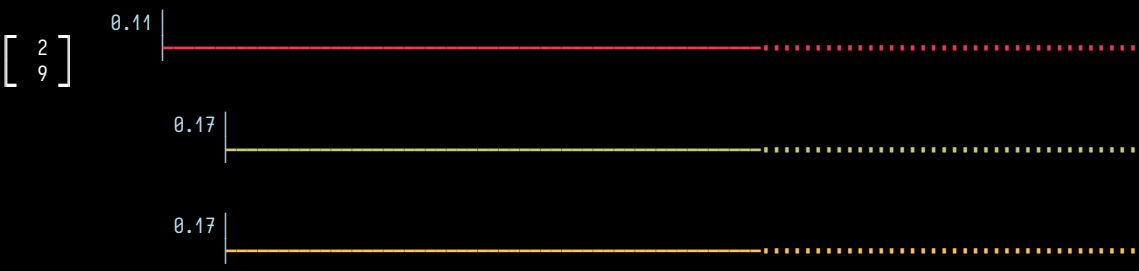
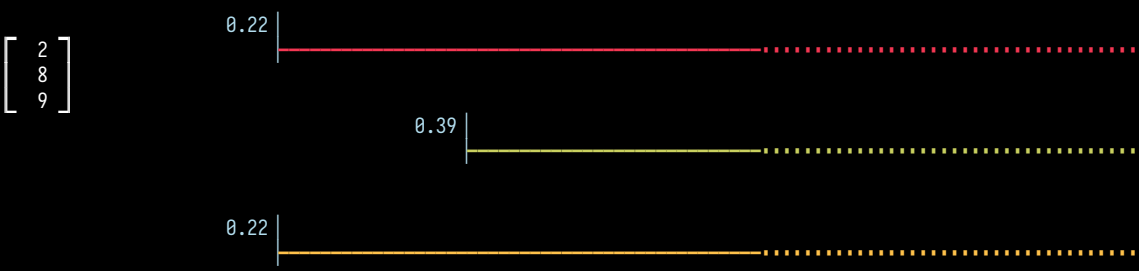
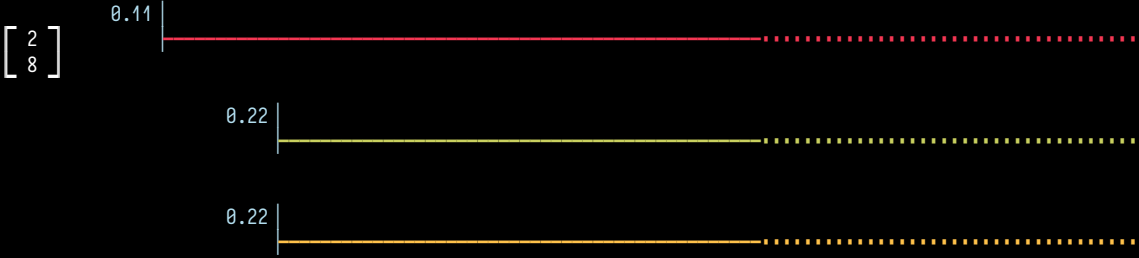


$\begin{bmatrix} 2 \\ 7 \\ 8 \\ 9 \end{bmatrix}$



$\begin{bmatrix} 2 \\ 7 \\ 9 \end{bmatrix}$





$$\begin{bmatrix} 5 \\ 7 \\ 8 \end{bmatrix}$$



$$\begin{bmatrix} 5 \\ 7 \\ 8 \\ 9 \end{bmatrix}$$



$$\begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$



$$\begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

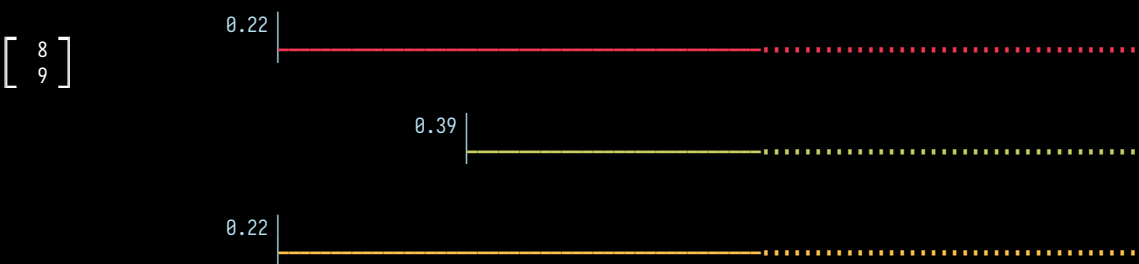
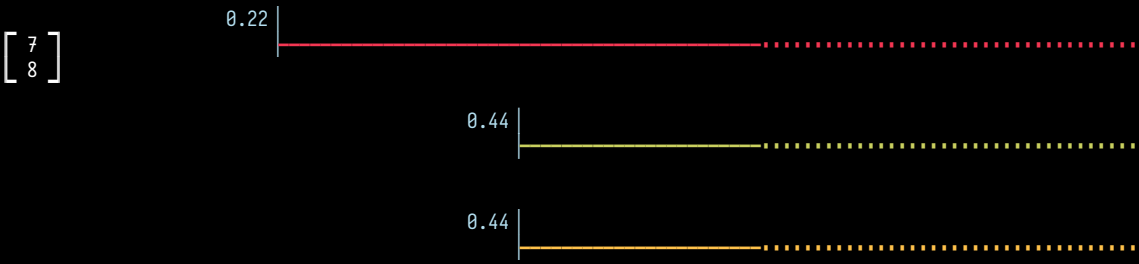
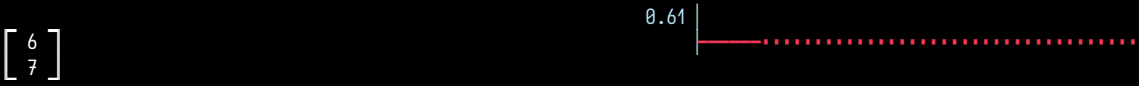


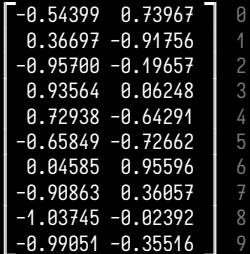
$$\begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix}$$



$$\begin{bmatrix} 5 \\ 9 \end{bmatrix}$$







∂_0
 $h_0=1$

Z^1_0 , 0-simplices: $[0]$ $[1]$ $[2]$ $[3]$ $[4]$ $[5]$ $[6]$ $[7]$ $[8]$ $[9]$

Z^{21} , 1-simplices: $\begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 8 \end{bmatrix}, \begin{bmatrix} 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \end{bmatrix}, \begin{bmatrix} 2 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix},$
 $\begin{bmatrix} 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 5 \\ 8 \end{bmatrix}, \begin{bmatrix} 5 \\ 9 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \end{bmatrix}, \begin{bmatrix} 7 \\ 9 \end{bmatrix}, \begin{bmatrix} 8 \\ 9 \end{bmatrix}$

$$h_2 = 0$$
$$\begin{bmatrix} 5 \\ 8 \end{bmatrix} \quad \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

[9], [9]

$\partial_3:$

$$\begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad h_3=0$$

Z^6 , 3-simplices:

$$\begin{bmatrix} 0 \\ 2 \\ 7 \\ 8 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 7 \\ 8 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 7 \\ 9 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 8 \\ 9 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 8 \\ 9 \end{bmatrix}, \begin{bmatrix} 5 \\ 7 \\ 8 \\ 9 \end{bmatrix}$$

$\partial_4:$

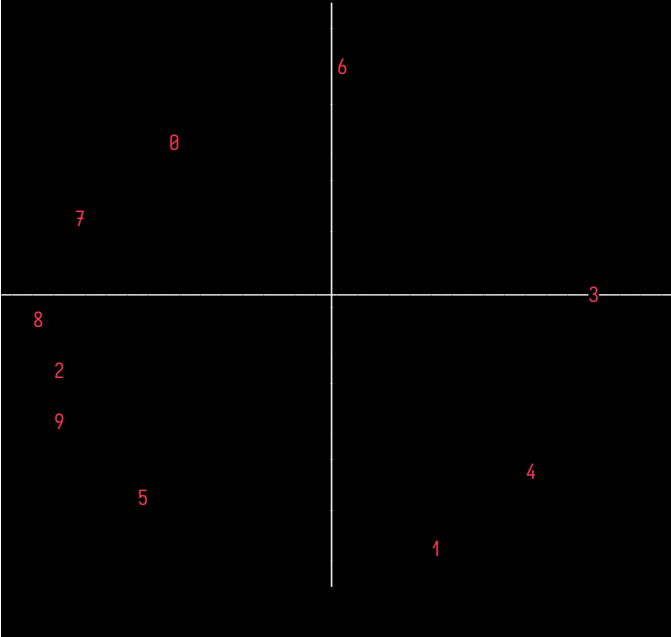
$$\begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} \quad h_4=0$$

Z^1 , 4-simplices:

$$\begin{bmatrix} 2 \\ 5 \\ 7 \\ 8 \\ 9 \end{bmatrix}$$

$\partial_5:$ [0] $h_5=0$

0



$$\begin{bmatrix} -0.54399 & 0.73967 \\ 0.36697 & -0.91756 \\ -0.95700 & -0.19657 \\ 0.93564 & 0.06248 \\ 0.72938 & -0.64291 \\ -0.65849 & -0.72662 \\ 0.04585 & 0.95596 \\ -0.90863 & 0.36057 \\ -1.03745 & -0.02392 \\ -0.99051 & -0.35516 \end{bmatrix} \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{matrix}$$

flag_complex<cech_complex<euclidean_space<double, 2>>> (ε=0.33000000000000007).

∂_0
 $h_0=3$

Z^{10} , 0-simplices: $\begin{bmatrix} 0 \end{bmatrix}$ $\begin{bmatrix} 1 \end{bmatrix}$ $\begin{bmatrix} 2 \end{bmatrix}$ $\begin{bmatrix} 3 \end{bmatrix}$ $\begin{bmatrix} 4 \end{bmatrix}$ $\begin{bmatrix} 5 \end{bmatrix}$ $\begin{bmatrix} 6 \end{bmatrix}$ $\begin{bmatrix} 7 \end{bmatrix}$ $\begin{bmatrix} 8 \end{bmatrix}$ $\begin{bmatrix} 9 \end{bmatrix}$

$$\begin{array}{c} \uparrow \\ \partial_1: \end{array} \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} \quad h_1=0$$

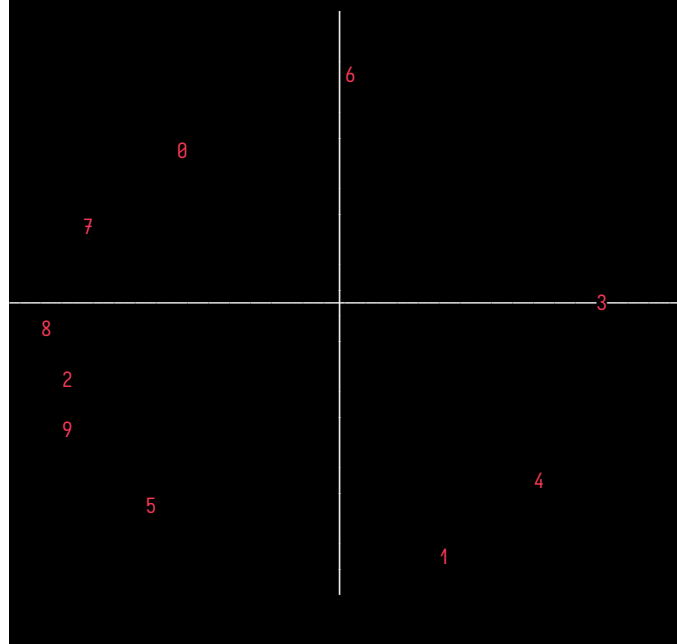
Z^{10} , 1-simplices: $\begin{bmatrix} 0 \\ 6 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 8 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 9 \end{bmatrix}$, $\begin{bmatrix} 5 \\ 9 \end{bmatrix}$, $\begin{bmatrix} 7 \\ 8 \end{bmatrix}$, $\begin{bmatrix} 8 \\ 9 \end{bmatrix}$

$$\begin{array}{c} \uparrow \\ \partial_2: \end{array} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad h_2=0$$

Z^3 , 2-simplices: $\begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 7 \\ 8 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 8 \\ 9 \end{bmatrix}$

$$\begin{array}{c} \uparrow \\ \partial_3: \end{array} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad h_3=0$$

0



$$\begin{bmatrix} -0.54399 & 0.73967 \\ 0.36697 & -0.91756 \\ -0.95700 & -0.19657 \\ 0.93564 & 0.06248 \\ 0.72938 & -0.64291 \\ -0.65849 & -0.72662 \\ 0.04585 & 0.95596 \\ -0.90863 & 0.36057 \\ -1.03745 & -0.02392 \\ -0.99051 & -0.35516 \end{bmatrix} \begin{array}{l} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{array}$$

$\alpha_shape_complex<euclidean_space<double, 2>> (\epsilon=0.6600000000000001)$

$$\begin{array}{c} \uparrow \\ \partial_0 \\ h_0=3 \end{array}$$

Z^{10} , 0-simplices: $\begin{bmatrix} 0 \end{bmatrix}$ $\begin{bmatrix} 1 \end{bmatrix}$ $\begin{bmatrix} 2 \end{bmatrix}$ $\begin{bmatrix} 3 \end{bmatrix}$ $\begin{bmatrix} 4 \end{bmatrix}$ $\begin{bmatrix} 5 \end{bmatrix}$ $\begin{bmatrix} 6 \end{bmatrix}$ $\begin{bmatrix} 7 \end{bmatrix}$ $\begin{bmatrix} 8 \end{bmatrix}$ $\begin{bmatrix} 9 \end{bmatrix}$

$$\begin{array}{c} \uparrow \\ \partial_1: \end{array} \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} \quad h_1=0$$

$$\mathbb{Z}^9, \text{ 1-simplices: } \begin{bmatrix} 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \end{bmatrix}, \begin{bmatrix} 2 \\ 9 \end{bmatrix}, \begin{bmatrix} 5 \\ 9 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \end{bmatrix}, \begin{bmatrix} 8 \\ 9 \end{bmatrix}$$

$$\begin{matrix} \uparrow \\ \partial_2: \end{matrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad h_2=0$$

$$\mathbb{Z}^2, \text{ 2-simplices: } \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \\ 9 \end{bmatrix}$$

$$\begin{matrix} \uparrow \\ \partial_3: \end{matrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad h_3=0$$

$$0$$

